Ultralightweight Feature-Compressed Multihead Self-Attention Learning Networks for Hyperspectral Image Classification

Xinhao Li, Mingming Xu, *Member, IEEE*, Shanwei Liu, Hui Sheng, and Jianhua Wan

***Abstract*— Vision transformers are widely used in hyperspectral image (HSI) classification, with their core feature extractor being self-attention. Self-attention has a wider receptive field than convolution. However, existing vision transformers for the classification of HSIs with a large number of bands generally suffer from high computational complexity and a large number of parameter requirements. In this article, we propose an ultralightweight feature-compressed multihead self-attention learning network (UFMS-LN), which mainly consists of a novel compressed feature multihead self-attention (CF-MHSA), a spatial feature enhancement-enhancing transformation reduction (SFE-ETR), and a spatial-spectral hybridization-receptive field attention convolutional operation (SH-RFAConv). By effectively compressing feature maps in spatial-spectral dimensions, CF-MHSA achieves the same feature extraction capabilities as state-of-the-art self-attention mechanisms, and its floating-point operations (FLOPs) and parameters are two orders of magnitude lower than state-of-the-art self-attention mechanisms. SH-RFAConv is designed to emphasize local features, which have the ability to extract both spatial-spectral features simultaneously and have a wider receptive field than traditional convolutional operations. Furthermore, SFE-ETR is a preprocessing module for UFMS-LN that combines global spatial feature enhancement methods with enhancing transformation reduction (ETR). Extensive experiments conducted on four benchmark HSI datasets have shown that this method achieves superior results compared to existing state-of-the-art HSI classification networks.**

***Index Terms*— Attention, deep learning, hyperspectral image (HSI) classification, remote sensing.**

I. Introduction

DUE to the abundant spectral information, hyperspectral images (HSIs) are widely used in various fields, such as environmental monitoring [1], marine exploration [2], and mineral analysis [3].

In recent years, deep learning [4] has been extensively applied to HSI classification [5], [6], [7], [8]. Due to the excellent feature extraction capabilities of convolutional kernels, convolutional neural networks (CNNs) can learn to extract more complex features by stacking multiple convolutional layers, leading to significant success in HSI classification [9], [10], [11]. In the early stages of research, 2-D CNNs [12], [13], [14] were initially attempted to extract deep semantic features. However, 2-D CNNs only focus on extracting spatial information and lack the ability to capture spectral information. In comparison, 3-D CNNs have the capability to extract both spatial and spectral features simultaneously [15], [16], [17]. Chen et al. [18] applied the deep feature extraction of 3-D CNNs to HSI classification, effectively addressing the issue of limited spectral information extraction in 2-D CNNs. Due to the high computational complexity of 3-D CNNs, Roy et al. [19] proposed a hybrid spectral CNN (HybridSN) that achieves better accuracy while reducing computational complexity. CNNs have powerful capabilities to capture local contextual information from HSIs due to the structural characteristics of convolutional kernels. However, the receptive field of CNNs is limited by the structure of the convolution kernel, which usually makes the network unable to model long-range dependencies. Additionally, CNNs commonly rely on stacking multiple convolutional layers to achieve deep feature extraction, which increases parameters and leads to high computational complexity [20].

Recently, vision transformers have been widely used in various fields of computer vision, such as image classification, object detection, and instance segmentation [21], [22], [23], [24]. Vision transformers primarily rely on self-attention mechanisms rather than convolutional operations for feature extraction. The self-attention mechanism possesses excellent capabilities for modeling long-range dependencies. By calculating the correlations between different positions, it can capture the nonlinear relationships among different elements in the input sequence. Therefore, the self-attention mechanism can effectively compensate for the inability of CNNs to model long-range dependencies in HSI classification [25], [26], [27], [28], [29], [30]. A novel multibranch transformer architecture addresses the issue of the network being susceptible to irrelevant information during training by incorporating spatial attention and extracting spectral features [31]. Furthermore, a multiscale feature extraction module is implemented in a transformer-based network to fully explore global and local multiscale features of pixels [32]. However, self-attention involves the operation of different matrices to model long-range dependencies, resulting in high computational costs [33].

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As the computational demands for HSI classification continue to increase, the research on lightweight classification models to strike a balance between high classification accuracy and low computational complexity has become increasingly important [34], [35], [36]. An end-to-end residual spectral-spatial attention network (RSSAN) utilizes spectral attention modules to emphasize relevant frequency bands and improves model training speed [37]. Compared to the existing band weighting module, the plug-and-play compact band weighting (CBW) module based on an attention-driven mechanism exhibits significantly fewer parameters, thereby reducing model complexity and time cost [38]. The LiteDepthwiseNet can achieve high classification performance with minimal parameters and excellent performance, which decomposes the standard convolution into depthwise convolution and pointwise convolution [39]. Although lightweight classification models have achieved good results, there are still some problems. First, self-attention in the transformer-based HSI classification framework has not made breakthrough progress in terms of computational complexity and parameter volume, and it still needs significant computational requirements. In addition, lightweight models effectively reduce computational requirements but may sacrifice some performance in classification.

To alleviate the above issue, we propose one solution: An ultralightweight feature-compressed multihead self-attention learning network (UFMS-LN). UFMS-LN mainly consists of four key components: spatial feature enhancement-enhancing transformation reduction (SFE-ETR), compressed feature multihead self-attention (CF-MHSA), and spatial-spectral hybridization-receptive field attention convolutional operation (SH-RFAConv). The purpose of constructing SFE-ETR is to achieve dimensionality reduction, enhance interclass differences, reduce intraclass differences, and simultaneously enhance the input of spatial features to provide sufficient features. The CF-MHSA module is designed to have an outstanding advantage in terms of computational complexity and parameters, showing a significantly reduced magnitude. Despite its lightweight size, it exhibits excellent feature extraction capabilities. In addition, the SH-RFAConv module possesses a wider spatial-spectral receptive field, which is crucial for local feature extraction. The main contributions of our work are as follows.

1) A UFMS-LN is proposed for HSI classification by compressing transformer-based multiscale feature maps. This network integrates global and local multiscale features for the classification of hyperspectral pixels.
2) A CF-MHSA with extremely low computational complexity and parameters is designed. Feature compression is applied to the spatial-spectral dimensions of the feature matrix, resulting in vectors that represent the spatial-spectral features. This approach effectively addresses the issue of high computational complexity caused by element-wise operations between matrix elements. Furthermore, the computational complexity is optimized by adjusting the order of operations. This further reduces the overall computational burden.
3) To overcome the limited receptive field of convolutional kernels and capture a broader spatial-spectral receptive field, we have designed SH-RFAConv. SH-RFAConv is a variant of the receptive field attention convolutional (RFAConv) operation that incorporates the ability to gather spatial-spectral dimension perspectives. This integration enables SH-RFAConv to achieve a more comprehensive receptive field.
4) We design a novel unsupervised dimensionality reduction method called SFE-ETR, which aims to enhance feature differences for subsequent better feature extraction. Building upon enhancing transformation reduction (ETR), a spatial feature branch is designed to extract spatial information through fixed stepwise band selection. This process aims to capture comprehensive spatial features and compensate for any loss of spatial information that may occur during dimensionality reduction.

The rest of the article is structured in the following way. In Section II, our proposed model is described in detail. In Section III, extensive experiments are performed on four HSI datasets to demonstrate the effectiveness of the proposed approach. Finally, Section IV draws relevant conclusions.

II. Proposed Method

A. Overall Framework

The proposed UFMS-LN model consists of a two-layer single-branch structure, with the feature map dimensions decreasing layer by layer (shown in Fig. 1). In this framework, SFE-ETR serves as the preprocessing module, CF-MHSA acts as the core operator for extracting global features, and SH-RFAConv functions as the module for extracting local features. Specifically, $\boldsymbol{I} \in \mathbf{R}^{H\times W\times D}$ represents the original HSI data, where $H \times W$ represents the spatial size of a patch of HSI and D represents the number of spectral bands. The preprocessed HSI is represented as $\boldsymbol{X} \in R^{H\times W\times C}$, where $C(C \ll D)$ is the number of spectral bands after preprocessing. $\boldsymbol{X}$ serves as the input feature map for the network and is processed through the SH-RFAConv module, CF-MHSA module L1, a fully connected layer that includes a linearization layer and an activation function Gaussian error linear unit (GELU), and CF-MHSA module L2. Certainly, it is worth noting that the fully connected layer is employed to reduce the dimensionality of the feature maps, while L1 and L2 are utilized to process feature maps of different dimensions.

B. CF-MHSA

The self-attention in traditional transformers can obtain a 2-D input matrix $\boldsymbol{X} \in \mathrm{R}^{s\times d}$ by compressing the spatial dimension of the input feature data. Where s and d represent the dimensions of $\boldsymbol{X}$, $s = H \times W$, and d represents the spectral dimension of the feature map. Taking a flattened feature map $\boldsymbol{X} \in R^{s\times d}$ as the input, a multihead self-attention (MHSA) block with M heads is formulated as

$$\boldsymbol{Q} = \boldsymbol{X}\boldsymbol{W}_Q,\ \boldsymbol{K} = \boldsymbol{X}\boldsymbol{W}_K,\ \boldsymbol{V} = \boldsymbol{X}\boldsymbol{W}_V \tag{1}$$

$$z^{(n)} = \sigma\left(\boldsymbol{Q}^{(n)}\boldsymbol{K}^{(n)}/\sqrt{t}\right)V^{(n)},\quad n = 1,\ldots,N \tag{2}$$

$$z = \mathrm{Concat}\{z^{(1)},\ldots,z^{(N)}\}\boldsymbol{W}_o. \tag{3}$$

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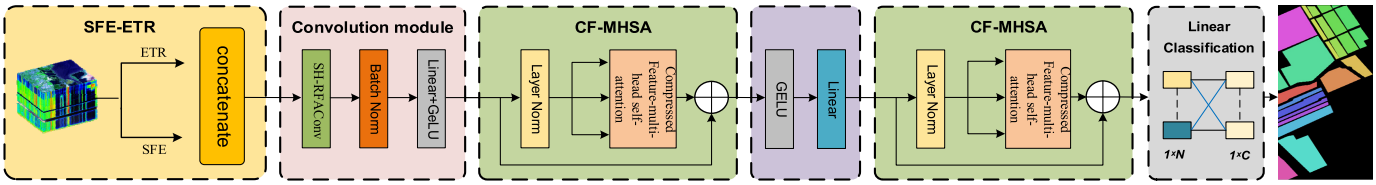

Fig. 1. Schematic representation of the proposed ultralightweight feature-compressed multihead self-attention learning network for HSI classification.

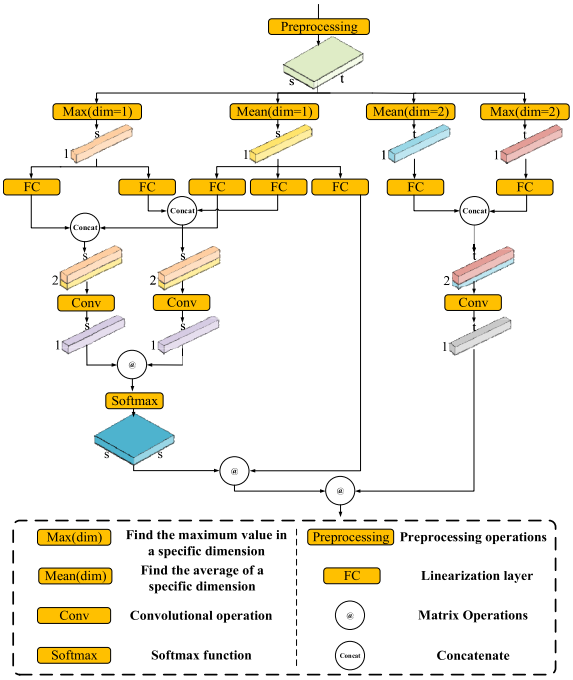

Fig. 2. CF-MHSA module.

Here, $\boldsymbol{W}_O$, $\boldsymbol{W}_Q$, $\boldsymbol{W}_K$, and $\boldsymbol{W}_V \in \mathbf{R}^{d\times d}$ are the projection matrices, and $\sigma(\cdot)$ denotes the softmax function. n denotes the number of heads, $t = d/n$ is the dimension of each head, and $z^{(n)}$ denotes the embedding output from the nth attention head. $\boldsymbol{Q}^{(n)}, \boldsymbol{K}^{(n)}, \boldsymbol{V}^{(n)} \in \mathrm{R}^{s\times t}$ denote query, key, and value embeddings, respectively. Concat$\{\cdot\}$ represents concatenation along the specified dimension.

MHSA applied to the global context can result in high computational costs because each position needs to consider its relationship with all other positions. To mitigate the problem, the proposed CF-MHSA greatly reduces the complexity and the number of parameters involved in calculations by compressing feature maps in different dimensions, as shown in Fig. 2.

Given the input data $\boldsymbol{X} \in \mathbf{R}^{s\times d}$, it can be viewed as $[\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_d]$, where $\boldsymbol{X}_i \in R^{s\times 1}$. In the first step, the maximum value $\max_i$, minimum value $\min_i$, and mean value mean_i for the ith vector are calculated. Then, the maximum and minimum values of each vector are replaced with mean_i as follows:

$$\begin{aligned}\max_i = \mathrm{Max}(\boldsymbol{X}_i);\ \min_i = \mathrm{Min}(\boldsymbol{X}_i);\ \mathrm{mean}_i = \mathrm{Mean}(\boldsymbol{X}_i)\\ -> \max_i = \mathrm{Mean}(\boldsymbol{X}_i);\ \min_i = \mathrm{Mean}(\boldsymbol{X}_i);\ \mathrm{mean}_i\\ = \mathrm{Mean}(\boldsymbol{X}_i)\end{aligned} \tag{4}$$

where Max$(\cdot)$, Min$(\cdot)$, and Mean$(\cdot)$ represent functions to obtain the maximum value, minimum value, and mean value, respectively. After performing the aforementioned operations, the matrix $\boldsymbol{X}' \in \mathbf{R}^{s\times d}$ is obtained and normalized $\boldsymbol{X}'$ using

$$\boldsymbol{X}' = \frac{\boldsymbol{X}' - \mathrm{Min}(\boldsymbol{X}')}{\mathrm{Max}(\boldsymbol{X}') - \mathrm{Min}(\boldsymbol{X}')}. \tag{5}$$

The purpose of the above steps is to replace any abnormally large or small values in the input data. This is done to prevent these outliers from affecting the extraction of feature vectors in subsequent steps. By replacing these outliers with more representative values, a more robust and reliable feature extraction can be ensured. Furthermore, all values in the feature map are nonnegative for subsequent tasks. Next, we divide $\boldsymbol{X}'$ along the spectral dimension into n parts, where n is the number of attention heads, $\boldsymbol{X}'_j \in \mathbf{R}^{s\times t}$, $t = d/n$, $j \in [1, N]$. For each $\boldsymbol{X}'_j$, the mean and maximum values are calculated separately for the spectral and spatial dimensions as follows:

$$\begin{aligned}\boldsymbol{max}_j^1 &= \mathrm{Max}(\boldsymbol{X}'|\mathrm{dim} = 1)\\ \boldsymbol{mean}_j^1 &= \mathrm{Mean}(\boldsymbol{X}'|\mathrm{dim} = 1)\\ \boldsymbol{max}_j^2 &= \mathrm{Max}(\boldsymbol{X}'|\mathrm{dim} = 2)\\ \boldsymbol{mean}_j^2 &= \mathrm{Mean}(\boldsymbol{X}'|\mathrm{dim} = 2).\end{aligned} \tag{6}$$

The vectors $\boldsymbol{max}_j^1 \in \mathbf{R}^{s\times 1}$, $\boldsymbol{max}_j^2 \in \mathbf{R}^{1\times t}$, $\boldsymbol{mean}_j^1 \in \mathbf{R}^{s\times 1}$, $\boldsymbol{mean}_j^2 \in \mathbf{R}^{1\times t}$ respectively represent the maximum and mean values of each head in different dimensions. Here, dim $= 1$ denotes the spatial dimension, and dim $= 2$ denotes the spectral dimension. The purpose of calculating these vectors is to prepare for feature compression by replacing the corresponding matrices with them in subsequent steps.

Subsequently, the vectors $(\boldsymbol{max}_j^1, \boldsymbol{mean}_j^1)$ and $(\boldsymbol{max}_j^2, \boldsymbol{mean}_j^2)$ are grouped, and each group is mapped separately, projecting them into a new space as follows:

$$\begin{aligned}\boldsymbol{max}_j^{1Q} &= \boldsymbol{max}_j^1 @ \boldsymbol{w}_q^1; \quad \boldsymbol{mean}_j^{1Q} = \boldsymbol{mean}_j^1 @ \boldsymbol{w}_q^2\\ \boldsymbol{max}_j^{1K} &= \boldsymbol{max}_j^1 @ \boldsymbol{w}_k^1; \quad \boldsymbol{mean}_j^{1K} = \boldsymbol{mean}_j^1 @ \boldsymbol{w}_k^2\\ \boldsymbol{mean}_j^{1V} &= \boldsymbol{mean}_j^1 @ \boldsymbol{w}_V^1\\ \boldsymbol{max}_j^{2V} &= \boldsymbol{max}_j^2 @ \boldsymbol{w}_V^2; \quad \boldsymbol{mean}_j^{2V} = \boldsymbol{mean}_j^2 @ \boldsymbol{w}_V^3\end{aligned} \tag{7}$$

where @ denotes matrix multiplication. $\boldsymbol{w}_q^1$, $\boldsymbol{w}_q^2$, $\boldsymbol{w}_k^1$, $\boldsymbol{w}_k^2$, $\boldsymbol{w}_V^1$, $\boldsymbol{w}_V^2$, and $\boldsymbol{w}_V^3 \in \mathbf{R}^{1\times 1}$ are the projection scalars for mapping $\boldsymbol{max}_j^-$, $\boldsymbol{mean}_j^-$ to different new spaces. Note that we use an $\mathbf{R}^{1\times 1}$ scalar instead of a $\mathbf{R}^{s\times s}$ or $\mathbf{R}^{t\times t}$ matrix for mapping in order to reduce the computational complexity. In addition, by mapping the 1-D vectors, the computational complexity is reduced to $O(s)$ and $O(d)$, replacing the mapping in the original self-attention mechanism with a complexity of $O(s^2 * (d/n))$.

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Next, we achieve $\boldsymbol{Q}^1$, $\boldsymbol{K}^1$, $\boldsymbol{V}^1$, and $\boldsymbol{V}^2$ by concatenating vectors that can represent a specific dimension of the original matrix and then compressing them using convolutional operations. The specific calculations are as follows:

$$\begin{aligned}\boldsymbol{Q}^1 &= f^{3\times1}\left(\text{Concat}\left\{\boldsymbol{max}_j^{1Q}, \boldsymbol{mean}_j^{1Q}\right\}\right)\\ \boldsymbol{K}^1 &= f^{3\times1}\left(\text{Concat}\left\{\boldsymbol{max}_j^{1K}, \boldsymbol{mean}_j^{1K}\right\}\right)\\ \boldsymbol{V}^1 &= \boldsymbol{mean}_j^{1V}\\ \boldsymbol{V}^2 &= f^{3\times1}\left(\text{Concat}\left\{\boldsymbol{max}_j^{2V}, \boldsymbol{mean}_j^{2V}\right\}\right)\end{aligned} \tag{8}$$

where $f^{3\times1}(\cdot)$ represents the convolution operation with a kernel size of 3×1; Concat$\{\cdot\}$ represents concatenation along the specified dimension, resulting in feature maps with sizes of $\mathbf{R}^{s\times2}$ and $\mathbf{R}^{2\times t}$; $\boldsymbol{Q}^1$, $\boldsymbol{K}^1$, and $\boldsymbol{V}^1 \in \mathbf{R}^{s\times1}$ and $V^2 \in \mathbf{R}^{1\times t}$ represent the calculated values of $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$, respectively. In order to increase the robustness of CF-MHSA and prevent the decline of feature extraction ability [40], we choose $\boldsymbol{max}_j^{1-}$ and $\boldsymbol{mean}_j^{1-}$ to obtain $\boldsymbol{Q}$ and $\boldsymbol{K}$ through processing, instead of $\boldsymbol{max}_j^{2-}$ and $\boldsymbol{mean}_j^{2-}$ with smaller parameter quantities. It is worth noting that the purpose of $\boldsymbol{V}^1$ not being compressed is to retain more features of the original feature matrix. Subsequently, $\boldsymbol{Q}^1$ and $\boldsymbol{K}^1$ is used to calculate attention scores. Due to both $\boldsymbol{Q}^1$ and $\boldsymbol{K}^1$ being 1-D feature compression vectors, the computational complexity of obtaining attention scores is only $O(s*(d/n))$

$$\boldsymbol{Attention} = \text{Softmax}(\boldsymbol{Q}^1@\boldsymbol{K}^1). \tag{9}$$

The softmax function, denoted as Sofmax$(\cdot)$, is used to obtain the attention scores. Finally, a new operation order for the compressed feature vector is proposed as follows:

$$\boldsymbol{Z}_{\text{out}} = (\boldsymbol{Attention}@\boldsymbol{V}^1)@\boldsymbol{V}^2 \tag{10}$$

where $\boldsymbol{Z}_{\text{out}}$ is the output feature map. Compared with $Z_{\text{out}} = \textit{Attention}@(\boldsymbol{V}^1@\boldsymbol{V}^2)$, the computational complexity is $O(s^2*(d/n)) = O(s*(s*(d/n))$, and the computational complexity of our proposed method is $O(s^2 + s*(d/n)) = O(s*(s+(d/n))$. In most cases, $(s+(d/n))$ is much smaller than $(s*(d/n))$.

C. SH-RFAConv

In order to enhance the spatial-spectral feature extraction capability of the UFMS-LN model, inspired by receptive field attention convolution operation (RFAConv) [41], we have designed a structure similar to RFAConv specifically for HSI features, called SH-RFAConv, as shown in Fig. 3. It has stronger spatial-spectral feature extraction capabilities by incorporating spatial-spectral attention.

In SH-RFAConv, a fast method called Group Convolution is utilized to extract both spatial and spectral features of the receptive field. The design of group convolution, which involves grouping the convolution operation, effectively reduces the number of parameters required and makes SH-RFAConv a lightweight method for capturing the interaction information. The number of groups in group convolution is set to the number of bands in the initial input feature map.

For SH-RFAConv, leveraging interactive receptive field feature information to learn attention maps can enhance network performance. SH-RFAConv can be divided into two parts: receptive-field feature map feature extraction and feature attention map acquisition. Let $\boldsymbol{X} \in \mathbf{R}^{B\times C\times H\times W}$ represent the input feature map, B denote the batch size, and C indicate the number of input bands, and H and W be the height and width of the feature map. On the one hand, we use Group Convolution $g^{3\times3}$ with a kernel size of 3×3 and group number set to C to extract feature map features, as follows.

$$\boldsymbol{F}_1 = \text{ReLU}\left(\text{Norm}\left(g_{C\times3\times3}^{3\times3}(\boldsymbol{X})\right)\right). \tag{11}$$

Here, $g_{C\times3\times3}^{3\times3}$ represents group convolution with a kernel size of 3×3 and output spectral size of $C\times3\times3$. $\boldsymbol{F}_1 \in \mathbf{R}^{B\times(C\times3\times3)\times H\times W}$ represents the obtained receptive-field feature map, Norm$(\cdot)$ indicates normalization and ReLU$(\cdot)$ represents the ReLU activation function. On the other hand, the feature attention map for the receptive field is obtained

$$\begin{aligned}\boldsymbol{A}_1 = \text{Softmax}\Big(&g_{C\times3\times3}^{1\times1}\left(g_C^{H\times W}(\boldsymbol{X})\right) * f_{3\times3}^{1\times1}\left(\text{AvgPool}_C^{3\times3}(\boldsymbol{X})\right)\\ &+ g_{C\times3\times3}^{1\times1}\left(\text{AvgPool}_C^{3\times3}(\boldsymbol{X})\right)\Big)\end{aligned} \tag{12}$$

$$\boldsymbol{F} = \boldsymbol{F}_1 + \boldsymbol{A}_1 \tag{13}$$

where $\text{AvgPool}_C^{3\times3}(\cdot)$ represents the average pooling operation, with a convolution kernel size of 3×3, and the output dimension is C. $f_{3\times3}^{1\times1}(\cdot)$ represents normal convolution with a kernel size of 1×1 and output spectral size of 3×3. The reason for using normal convolution is that the input and output dimension requirements are more flexible and are not affected by the number of groups. $\boldsymbol{A}_1 \in \mathbf{R}^{B\times(C\times3\times3)\times H\times W}$ represents the feature attention map. In operation $\boldsymbol{A}_1$, the features obtained sequentially from left to right are the attention map of the spectral dimension $\boldsymbol{A}_1^1 \in \mathbf{R}^{B\times(C\times3\times3)\times1\times1}$, the attention map of the spatial dimension $\boldsymbol{A}_1^2 \in \mathbf{R}^{B\times(1\times3\times3)\times H\times W}$, and the global attention map $\boldsymbol{A}_1^3 \in \mathbf{R}^{B\times(C\times3\times3)\times H\times W}$. Through (11), we can find that $\boldsymbol{A}_1^1$ and $\boldsymbol{A}_1^2$ are obtained by performing similar Channel attention [42] and spatial attention processing on $\boldsymbol{X}$, respectively, providing spatial-spectral attention for SH-RFAConv. After $\boldsymbol{A}_1^1$ and $\boldsymbol{A}_1^2$ are operated on each other and added to $\boldsymbol{A}_1^3$, the feature attention map is obtained through the Softmax function. It is worth noting that during the design process of fusion of $\boldsymbol{A}_1^1$ and $\boldsymbol{A}_1^2$ feature maps, we found through experiments that the multiplication fusion method has better results, so we chose to use the multiplication method. $\boldsymbol{F}$ is obtained by adding the attention map $\boldsymbol{A}_1$ with the feature map $\boldsymbol{F}_1$. Notably, the feature map $\boldsymbol{F} \in \mathbf{R}^{B\times(C\times3\times3)\times H\times W}$ obtained through SH-RFAConv, after being "reshaped," do not overlap, allowing the learned attention map to aggregate all the feature information of the feature map. After reshaping, the feature map $\boldsymbol{F} \in \mathbf{R}^{B\times C\times(3\times H)\times(3\times W)}$ needs to undergo a 3×3 convolution operation with a step size of 3 to achieve nonoverlapping feature extraction. The specific operations are as follows:

$$\boldsymbol{Z}_{\text{out}} = \text{ReLU}\left(\text{Norm}\left(f_{C1}^{3\times3}(\boldsymbol{F})\right)\right) \tag{14}$$

where $\boldsymbol{Z}_{\text{out}} \in \mathbf{R}^{B\times C1\times H\times W}$ is the output feature map and $C1$ is the spectral dimension of the output.

SH-RFAConv can effectively capture information differences caused by spatial variations, achieved by emphasizing the importance of different features within the receptive field

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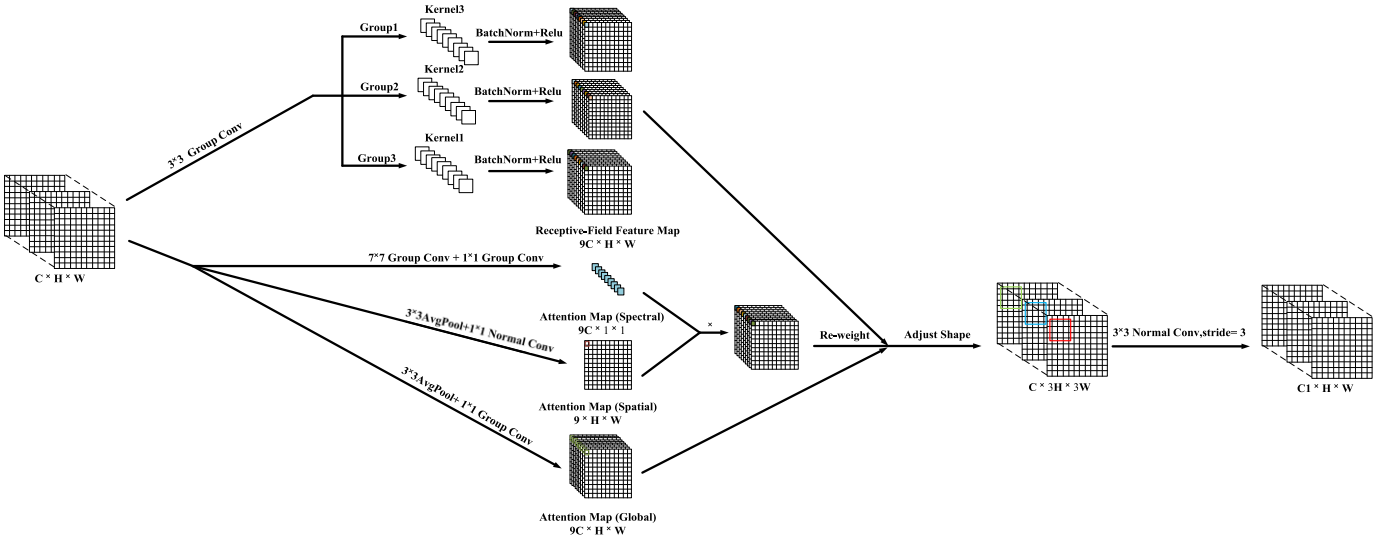

Fig. 3. Detailed structure of spatial-spectral hybridization-receptive field attention convolutional operation.

$\boldsymbol{A}_1$ and fusing it with the feature map $\boldsymbol{F}_1$. At the same time, compared to RFAConv, SH-RFAConv possesses a broader range of receptive field scales and stronger attention capabilities.

D. SFE-ETR

ETR [43] is a novel preprocessing method that reduces the dimensionality of input data by transferring the primary feature information to another subspace, enhancing the transformation and making the data smoother. However, ETR lacks the capability to extract spatial features, facing the challenge of significant loss of spatial features after dimensionality reduction. Therefore, we have augmented the ETR method by incorporating a spatial information approach with summarization capabilities, efficiently extracting spatial features, named SFE-ETR. The SFE-ETR structure is a dual-branch processing structure, which respectively plays the role of dimensionality reduction, distribution enhancement, and spatial feature enhancement.

ETR can be divided into two steps: dimensionality reduction and enhanced transformation. In the first step, we take the primary input dataset $X_{b\times q}$, where b represents the number of pixels and q represents the number of bands. The covariance matrix calculates the distance values based on the mean values of each variable, thereby enhancing the variation between input features, as shown in the following equation:

$$\boldsymbol{C}_x = \left(\frac{\left((\boldsymbol{X} - \bar{X})(\boldsymbol{X} - \bar{X})^T \right)}{b-1} \right) - \boldsymbol{\varepsilon} \tag{15}$$

$$\boldsymbol{W}_x = \text{linal}g_{\text{eig}}(\boldsymbol{C}_x) \tag{16}$$

where $\bar{\boldsymbol{X}} \in \mathbf{R}^{b\times 1}$ is the mean value of $\boldsymbol{X}$; $\boldsymbol{C}_x$ is the enhanced covariance matrix, $\boldsymbol{C}_x \in \mathbf{R}^{q\times q}$; and $\boldsymbol{\varepsilon} \in \mathbf{R}^{q\times q}$ are tiny values belonging to the range $[-1, 1]$. They are calculated from the Pearson correlation coefficient [44], which is affected by the correlation between pixels. In simple terms, if the correlation coefficient is high, the value of ε is large, and vice versa. $\text{linal}g_{\text{eig}}$ refers to linear feature decomposition, used to calculate the feature matrix $\boldsymbol{W}_x \in \mathbf{R}^{q\times q}$. Next, we take the vector weight matrix ($\boldsymbol{W} \in \mathbf{R}^{q\times k}$) of the first k dimensions from $\boldsymbol{W}_x$. Finally, the final new subspace dimensions, which extract the features of the primary input dataset, are obtained by multiplying $\boldsymbol{W}$ with $\boldsymbol{X}$. It is worth noting that k is generally less than q to achieve dimensionality reduction, as shown below

$$\tilde{\boldsymbol{\Gamma}}_{b\times k} = \boldsymbol{X}_{b\times q} \times (\boldsymbol{W}_{q\times k})^T \tag{17}$$

where $\tilde{\boldsymbol{\Gamma}}_{b\times k}$ is $\boldsymbol{X}$ being mapped to the new subspace data.

The next step is the enhanced transformation. This part yields the output of the morphological reconstruction method (MRM). MRM primarily consists of morphological dilation and erosion reconstruction, and this study employs the morphological dilation method. Morphological dilation operation (MDO) depends on two images (datasets), the mask ($\boldsymbol{S}$) and the marker ($\boldsymbol{\mathcal{R}}$) image, which must have the same size. The output from the previous part serves as the mask image. The main challenge in this part lies in obtaining the marker image. In addition, during the MDO, the values of the marker image must be less than or equal to the values of the mask image, i.e., $\boldsymbol{\mathcal{R}} \leq \boldsymbol{S}$.

Let $\tilde{\boldsymbol{\Gamma}}_{b\times k}$ be the primary input data of the second part, where b represents the number of instances and k represents the number of features (bands). This study enhances the variation within the class by using the average or maximum values of instances in this part. Therefore, before creating the $\boldsymbol{\mathcal{R}}$ image, two states are present

$$\boldsymbol{\mathcal{R}} = \begin{cases} \sum_{b=1}^{b} \tilde{\boldsymbol{\Gamma}}_b / b \times \sigma, & \text{if } \boldsymbol{\mathcal{R}} \leq \boldsymbol{S} \\ \text{argmax}\left(\tilde{\boldsymbol{\Gamma}}_b\right) \times \sigma, & \text{if } \boldsymbol{\mathcal{R}} > \boldsymbol{S} \end{cases} \tag{18}$$

where $\boldsymbol{\mathcal{R}}$ is the created marker image and σ is a constant value used to scale the values in the $\boldsymbol{\mathcal{R}}$ image. Since MDO is chosen, it is required that $\boldsymbol{\mathcal{R}} \leq \boldsymbol{S}$. The average value of each column in $\boldsymbol{S}$ resulting in $\boldsymbol{\mathcal{M}}_b \in \mathbf{R}^{b\times 1}$ is calculated. Then, we repeat

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the vector $\boldsymbol{\mathcal{M}}_b k$ times to transform the 1-D vector ($\boldsymbol{\mathcal{M}}_b$) into a 2-D matrix $\boldsymbol{\mathcal{R}} \in \mathbf{R}^{b\times k}$.

After obtaining the marker image, the process of obtaining the transmitted image using MDO is defined as follows:

$$D_S^{\delta}(\boldsymbol{\mathcal{R}}) = \delta_S^n(\boldsymbol{\mathcal{R}}) = \delta_S^1\left(\delta_S^{n-1}(\boldsymbol{\mathcal{R}})\right) \Lambda \boldsymbol{S} \tag{19}$$

$$\widehat{\boldsymbol{X}} = \text{normalize_gaussian}\left(D_S^{\delta}(\boldsymbol{\mathcal{R}})\right) \tag{20}$$

where $\boldsymbol{S}$ represents the mask image, $\boldsymbol{S} \geq \boldsymbol{\mathcal{R}}$, $n \geq 1$; δ represents MDO Λ represents the point-wise minimum operation; normalize_gaussian$(\cdot)$ represents the Gaussian normalized operation; and $\widehat{\boldsymbol{X}} \in \mathbf{R}^{b\times k}$ represents the output feature map of branch 1.

In multiobjective optimization, the $\boldsymbol{\mathcal{R}}$ image is used to label regions and emphasize the relevant values for each class in the $\boldsymbol{S}$ image, thereby enhancing the transformation. In other words, the transformation of the $\boldsymbol{\mathcal{R}}$ is controlled and constrained by the $\boldsymbol{S}$ in MDO. The peaks in the $\boldsymbol{\mathcal{R}}$ act as seed values that expand to fill the $\boldsymbol{S}$. MDO helps reduce noise values and correct the position of each value to move it closer to its respective class. It is worth noting that the "enhanced transformation" mentioned is to enhance the change of each pixel to the category it belongs to and to increase the difference between pixels in different categories. The last step in the second part is to normalize the output using a Gaussian distribution. This process aids in further reducing outliers and improves classification performance.

In the spatial feature enhancement branch, we have designed a method for fixed-step band extraction and perform feature extraction on each extracted band individually.

$\boldsymbol{X}_{b\times q}$ represents the input data, where b represents the number of instances and q represents the number of spectral bands. By partitioning along the spectral band dimension, $X_{b\times q}$ can be regarded as consisting of q $\boldsymbol{X}_{b\times 1}$. Band extraction is performed using a fixed step size t and the general formula is as follows:

$$\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_i, \ldots, \boldsymbol{x}_q\} \tag{21}$$

$$\tilde{\boldsymbol{X}} = \{\boldsymbol{x}_1, \boldsymbol{x}_{1+t}, \boldsymbol{x}_{1+2t}, \ldots, \boldsymbol{x}_{1+nt}\} \tag{22}$$

where $\boldsymbol{x}_i \in \mathbf{R}^{b\times 1}$ represents the image of the ith spectral band in the HSI, $1 \leq i \leq q$ and n is the total number of movements from the first band to the last band within the fixed step size, t. $\tilde{\boldsymbol{X}}$ is a summary data of the original data, which contains spatial features that can represent the spatial characteristics of the original data, but with fewer spectral bands. Finally, spatial feature extraction is performed on $\tilde{\boldsymbol{X}}$ band by band. The local binary pattern (LBP) method is an operator used to describe local features in an image. LBP possesses significant advantages such as grayscale invariance and rotation invariance, making them widely applied in various computer vision fields.

LBP is adopted to extract spatial information and the general formula is as follows:

$$\tilde{\boldsymbol{X}}_j = \text{Local_binary_pattern}\left(\tilde{\boldsymbol{X}}_j\right) \tag{23}$$

where Local_binary_pattern$(\cdot)$ represents the operation of a local feature operator on the image; $j \in [1, n]$; $\tilde{\boldsymbol{X}} \in \mathbf{R}^{b\times(1+n)}$ represents the feature map output from branch 2. Finally, we concatenate the feature maps $\widehat{\boldsymbol{X}}$ and $\tilde{\boldsymbol{X}}$ along the spatial dimension to generate the output of the SFE-ETR, which is denoted as $\boldsymbol{Output} \in \mathbf{R}^{b\times(1+nt+k)}$

$$\boldsymbol{Output} = \text{Concat}\{\hat{\boldsymbol{X}}, \tilde{\boldsymbol{X}}\}. \tag{24}$$

Ground Truth	No.	Color	Name	Train	Val	Test
Pavia University	1		Asphalt	66	66	6499
	2		Meadows	186	187	18276
	3		Gravel	21	21	2057
	4		Trees	30	31	3003
	5		Paintedmetalsheets	13	14	1318
	6		BareSoil	50	50	4929
	7		Bitumen	14	13	1303
	8		Self-BlockingBricks	37	37	3608
	9		Shadows	10	9	928
	Total			427	428	41921
WHU-LongKou	1		Corn	172	173	34166
	2		Cotton	42	42	8290
	3		Sesame	15	15	3001
	4		Broad-leafsoybean	316	316	62580
	5		Narrow-leafsoybean	21	21	4109
	6		Rice	59	60	11735
	7		Water	335	335	66386
	8		Roadsandhouses	36	35	7053
	9		Mixedweed	26	26	5177
	Total			1022	1023	202497
Salinas	1		Brocoli_green_weeds_1	20	20	1969
	2		Brocoli_green_weeds_2	37	37	3652
	3		Fallow	20	20	1936
	4		Fallow_rough_plow	14	14	1366
	5		Fallow_smooth	27	27	2624
	6		Stubble	40	39	3880
	7		Celery	36	36	3507
	8		Grapes_untrained	112	113	11046
	9		Soil_vinyard_develop	62	62	6079
	10		Corn_senesced_green_we	33	33	3212
	11		Lettuce_romaine_4wk	10	11	1047
	12		Lettuce_romaine_5wk	20	19	1888
	13		Letuce_romaine_6wk	9	9	898
	14		Lettuce_romaine_7wk	11	10	1049
	15		Vinyard_untrained	72	73	7123
	16		Vinyard_vertical_trellis	18	18	1771
	Total			541	541	53047
Indian Pines	1		Alfalfa	10	3	33
	2		Corn-notill	43	15	1370
	3		Corn-mintill	25	9	796
	4		Corn	10	3	224
	5		Pasture	15	5	463
	6		Trees/Grass	22	8	700
	7		Pasture-mowed	10	3	16
	8		Hay-windrowed	15	5	458
	9		Oats	10	3	8
	10		Soybean-notill	30	10	932
	11		Soybean-mintill	74	25	2356
	12		Soybean-cleantill	18	6	570
	13		Wheat	10	3	192
	14		Woods	38	13	1214
	15		Buildings-Grass	12	4	370
	16		Stone-SteelTowers	10	3	80
	Total			352	118	9782

Fig. 4. Detailed information of four datasets for experiments.

III. Experiments and Result Analysis

A. Datasets

In the experiments, we use four HSI datasets to evaluate the effectiveness of the proposed method, including WHU-LongKou (WHU-LK), Pavia University (PU), Indian Pines (IP), and Salinas (SA), as shown in Fig. 4.

1) *Salinas:* The Salinas dataset was acquired by the AVIRIS Imaging Spectrometer in the Salinas Valley, California, and covers an area of 512 × 217 pixels with a spatial resolution of 3.7 m and contains 16 different feature coverage classes. The Salinas dataset has a total of 224 spectral bands, with 20 absorption bands removed, leaving 204 spectral bands for experimental use.
2) *Pavia University:* The PU dataset was acquired in 2003 by the reflectance optical system for imaging spectroscopy (ROSIS) on board at the University of Pavia, Italy, with a spatial resolution of 1.3 m, comprising 610 × 340 pixel points and nine classes. This PU dataset has 115 spectral bands, of which 12 are absorbing and noisy, and only 103 bands were used for experimental tests.
3) *Indian Pines:* The hyperspectral dataset was acquired by the airborne visible/infrared imaging spectrometer

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(AVIRIS) sensor over the northwest region of Indiana, USA. The scene consists of 220 spectral bands in the wavelength range of 400 to 2500 nm with a spectral resolution of 10 nm, a ground sampling distance (GSD) of 20 m, and a size of 145 × 145 pixels. By removing 20 noise and water absorption bands, i.e., bands 104–108, 150–163, and 220, we retained 200 channels.

4) *WHU-LongKou:* The WHU-LongKou dataset was captured in Longkou City, Hubei Province, using a nanoscale hyperspectral imaging sensor with a focal length of 36 mm, an image size of 400 × 270 pixels, and a spatial resolution of about 463.1 m. The WHU-LongKou dataset has 109 bands with wavelengths ranging from 400 to 1000 nm.

Each dataset was assigned a different proportion of labeled samples. The number of each class for training, validation, and testing sets of four datasets can be found in Fig. 4.

B. Experimental Setups

To better evaluate the experimental results, we use overall accuracy (OA), average accuracy (AA), and Kappa coefficient to assess the final classification outcomes on the four datasets. OA represents the proportion of correctly classified pixels to the total number of pixels, providing an overall measure of classification accuracy. AA corresponds to the average accuracy of each class in all categories, reflecting the overall performance across different types and indicating the stability of the classification results. Kappa coefficient is used to assess the consistency and effectiveness of the classification.

To ensure reliability, the labeled samples were split five times, and all reported results are the averages of the five repetitions. Standard deviation is also calculated to provide a measure of result variability, enhancing the credibility of the outcomes. Additionally, we analyze the complexity of the model using the number of parameters (Params) and floating-point operations (FLOPs), which provide insights into the computational requirements of the model. All experiments were conducted on an NVIDIA RTX 3060 GPU with 12 GB of memory. To ensure fair performance comparison between different models, all classification methods were implemented within the PyTorch framework. The training for all models was conducted for 350 epochs to update all model parameters. The UFMS-LN model is composed of two layers, with feature map dimensions set as {30, 10}, respectively. In order to obtain the best performance of other models participating in the comparison, the optimizer and learning rate scheduler were kept consistent with the original settings.

C. Parameter Analysis

1) Comparison of Learning Rates and Batch Sizes: Fig. 5 shows the OA under different learning rates and batch size parameters. The regions with higher classification accuracy are mainly located on the right side of the graph. Additionally, the influence of batch size and learning rate on the classification accuracy of the UFMS-LN method is relatively small for the WHU-LK dataset. In this study, the learning rate was set to 0.01. The minimum batch size was chosen as 50 for the SA, IP, and PU datasets and 100 for the WHU-LK dataset.

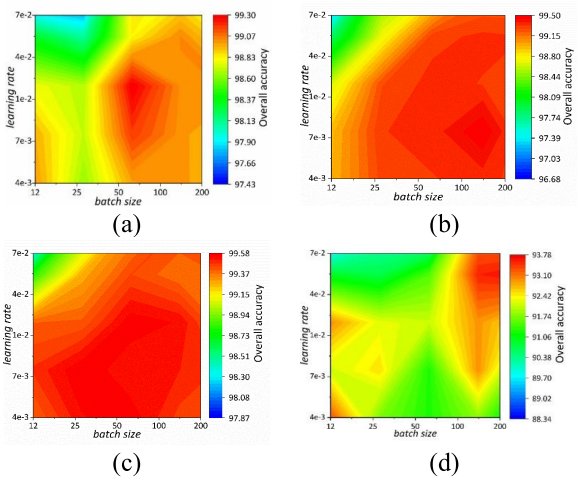

Fig. 5. Performance of UFMS-LN with different learning rates and batch size on the four datasets. (a) SA. (b) PU. (c) WHU-LK. (d) IP.

TABLE I
EFFECT OF PHASE DIMENSION ASSIGNMENT ON OA

120	60	30	20	10	SA	PU	WHU-LK	IP
√	√				99.41	99.23	99.49	92.46
√		√			99.41	99.37	99.56	91.76
√			√		99.51	99.23	99.51	92.51
√				√	99.50	99.30	99.53	90.18
	√	√			99.33	99.22	99.52	91.66
	√		√		99.40	99.14	99.53	90.94
	√			√	99.44	99.21	99.52	90.43
		√	√		99.27	99.20	99.57	91.46
		√		√	99.39	99.30	99.57	92.11
			√	√	99.36	99.11	99.50	89.14

2) Comparison of Different Feature Map Dimensions: In order to obtain the optimal feature map dimensions, relevant experiments were conducted, as shown in Table I. We find that when the dimensions are set to {30, 10}, the classification obtained has excellent accuracy and the number of feature map dimensions remains low on the four datasets.

3) Size of Spatial Input of HSIs: In this section, experiments on six patch sizes with spatial sizes {3, 5, 7, 9, 11, 13} were conducted, which were compared on four datasets. As shown in Fig. 6, the classification accuracy does not always improve with increasing patch size, and on the PU dataset, the accuracy instead decreases when the size exceeds 7 × 7. The main reason for this phenomenon is the different sizes of smoothing regions in different datasets, with the SA dataset having a larger smoothing region and the PU dataset having a smaller smoothing region. Meanwhile, on all four datasets, the accuracy changes less when the size exceeds 7 × 7. Therefore, we chose the 7 × 7 patch for our experiments.

4) Comparison of SFE-ETR Dimension Variations: SFE-ETR, as the preprocessing module of this model, consists of two main parts: spatial information processing and ETR. An accuracy comparison experiment of the different output data dimensions of the two parts of SFE-ETR was conducted.

As shown in Fig. 7, the effect of different dimensions on the model accuracy was evaluated by adjusting the dimension settings of the two parts. We observe that the accuracy of

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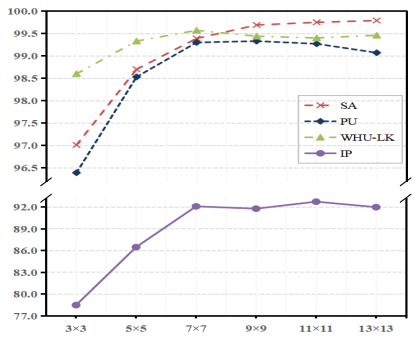

Fig. 6. OAs (%) with different spatial sizes on the four datasets.

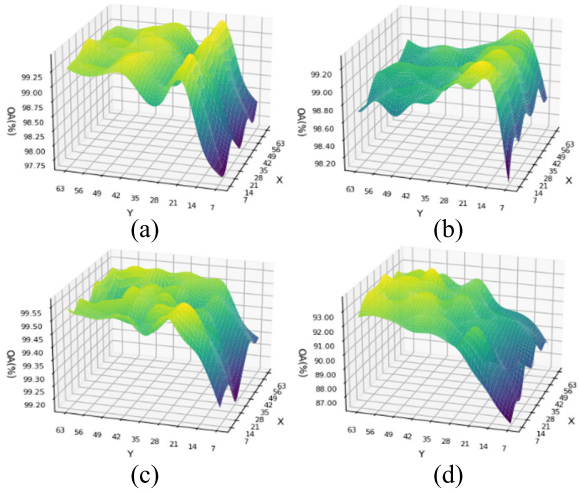

Fig. 7. OAs (%) with output dimensions for different parts of the SFE-ETR. (a) SA. (b) PU. (c) WHU-LK. (d) IP. The y-axis corresponds to the values of the ETR output dimensions and the x-axis corresponds to the values of the spatial information processing part of the output dimensions.

the model reaches a high level when the output dimension of spatial information processing is set to 14 on both SA and PU datasets. At the same time, it was found that when the output dimension of spatial information processing is set to 14, and the output dimension of ETR is set to 21, the model accuracy reaches a local maximum on the different datasets, and the combined overall dimensions are less. Therefore, we uniformly chose to set the output dimension of spatial information processing to 21, the output dimension of ETR to 14, and the merged overall dimension of 35 bands as inputs to the model.

D. Comparison With Other Models

To verify the model's superiority, a series of experiments were conducted to compare the results of the proposed method with several state-of-the-art deep learning methods. The comparison models are a multiscale 3-D deep CNN (M3D-DCNN) [45], a CNN-based 3-D DL approach (3D-CNN) [16], a deep feature fusion network (DFFN) [46], a residual spectral–spatial attention network [37], an attention-based bidirectional long short-term memory network (AB-LSTM) [47], a spectral–spatial feature tokenization transformer (SSFTT) [48], a group-aware hierarchical transformer (GAHT) [49], a dual-view spectral and global spatial feature fusion network (DSGSF) [50], and graph generative structure-aware Transformer (GraphGST) [51] for HSIs classification.

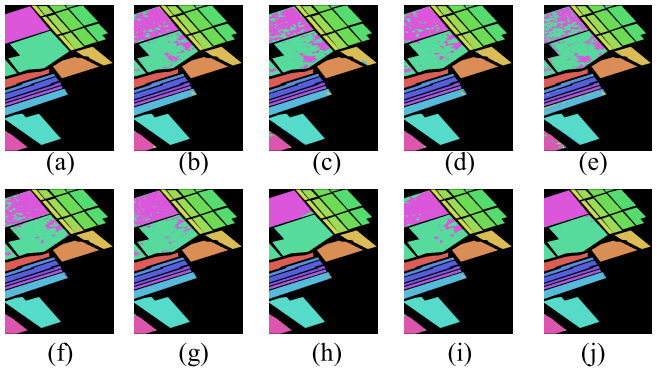

Fig. 8. Classification maps of the SA dataset with 1% training samples. (a) Ground truth. (b) M3D-DCNN. (c) 3D-CNN. (d) DFFN. (e) AB-LSTM. (f) SSFTT. (g) GAHT. (h) DSGSF. (i) GraphGST. (j) Proposed UFMS-LN.

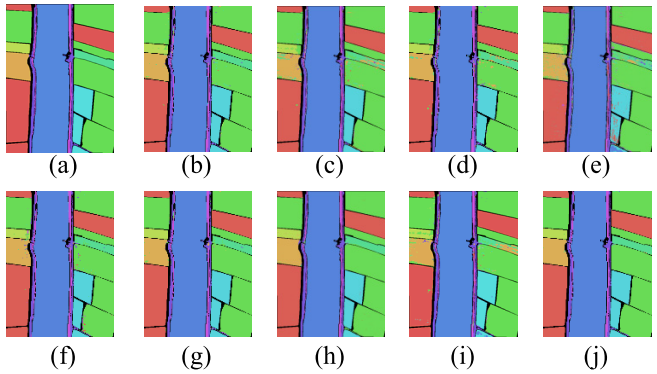

Fig. 9. Classification maps of the WHU-LK dataset with 0.5% training samples. (a) Ground truth. (b) M3D-DCNN. (c) 3D-CNN. (d) DFFN. (e) AB-LSTM. (f) SSFTT. (g) GAHT. (h) DSGSF. (i) GraphGST. (j) Proposed UFMS-LN.

Tables II–V show the classification results of the different methods. The results show that on the four benchmark datasets, UFMS-LN outperforms other deep learning-based classification methods in almost every metric. For example, on the SA dataset, the OA of UFMS-LN is 0.24% higher than DSGSF, 2.62% higher than GAHT, and 5.39% higher than SSFTT. In addition, UFMS-LN also performs better in terms of AA and Kappa values compared to other models. Similar results are observed on the WHU-LK dataset, where UFMS-LN outperforms the other models in terms of OA, AA, and Kappa values. On the PU dataset, UFMS-LN showed dominance in most of the categories and showed significant dominance in all four evaluation metrics. On the Indian Pines dataset, UFMS-LN slightly underperforms the GraphGST model in OA and Kappa metrics. However, combining the four datasets, GraphGST is much less accurate than UFMS-LN on the SA, WHU-LK, and PU datasets. Overall, the results of the study show that UFMS-LN achieves significant improvements in classification compared to other deep learning-based methods on the evaluated datasets.

The classification maps are shown in Figs. 8–11. The classification results of M3D-DCNN and 3D-CNN contain a significant amount of noise. When observing all the classifica-

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TABLE II

CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE PU DATASET WITH 1% TRAINING SAMPLES

Class	M3D-DCNN	3D-CNN	DFFN	AB-LSTM	SSFTT	GAHT	DSGSF	GraghGST	Proposed
1	96.61± 0.86	93.37± 1.52	96.81 ± 1.01	92.71 ± 0.93	96.91 ± 1.44	96.77 ± 1.98	97.60±0.99	97.19±0.54	**99.46 ± 0.50**
2	97.47 ±1.03	97.98± 0.50	99.55 ± 0.27	98.47 ± 0.76	99.33 ± 0.41	99.46 ± 0.18	99.20±0.37	99.10±0.40	**99.97 ± 0.03**
3	80.61 ± 6.69	61.01± 6.95	78.33 ± 7.15	69.96 ± 10.48	83.77 ± 8.07	85.25 ± 6.82	**96.73±0.86**	95.28±3.03	96.26 ± 0.57
4	97.76 ± 1.02	96.63± 0.83	95.40 ± 1.72	91.52 ± 2.58	96.94 ± 1.81	96.70 ± 1.76	**98.99±0.31**	98.73±0.36	97.86 ± 0.47
5	99.68 ± 0.64	99.24± 0.96	99.88 ± 0.17	99.15 ± 0.83	99.98 ± 0.03	100.00 ± 0.00	99.89±0.07	98.90±1.89	**100.00 ± 0.00**
6	89.12 ± 1.52	64.35± 6.98	95.69 ± 2.65	84.36 ± 3.49	98.49 ± 0.90	98.18 ± 0.84	98.95±0.69	97.67±1.72	**99.99 ± 0.02**
7	83.82 ± 4.52	66.51± 2.80	85.82 ± 8.67	77.87 ± 5.97	91.27 ± 3.86	94.01 ± 3.26	95.87±3.39	97.05±1.03	**99.26 ± 1.10**
8	91.08 ± 4.16	90.97± 3.07	92.77 ± 6.01	83.33 ± 4.75	95.48 ± 1.62	95.31 ± 2.01	**97.96±1.14**	90.41±0.71	97.39 ± 0.86
9	99.33 ± 0.56	98.30± 0.88	99.42 ± 0.59	98.00 ± 1.82	98.99 ± 1.60	**99.57 ± 0.30**	98.86±1.00	99.39±0.44	99.20 ± 0.74
OA (%)	94.68 ± 0.97	89.87± 1.31	96.33 ± 1.05	92.09 ± 1.36	97.35 ± 0.42	97.49 ±0.51	98.59± 0.21	97.57±0.17	**99.30 ± 0.14**
AA (%)	92.83 ± 1.47	85.37± 1.63	93.74 ± 1.72	88.37 ± 1.78	95.68 ± 0.97	96.14 ± 0.72	98.23± 0.45	97.08±0.65	**98.82 ± 0.20**
K×100	92.95 ± 1.27	86.35± 1.80	95.12 ± 1.40	89.43 ± 1.82	0096.49 ±	96.67 ± 0.68	98.13± 0.28	96.78±0.22	**99.07 ± 0.19**
Train time (s)	56.6	55.54	106.54	79.39	67.2	71.71	345.03	59.22	**53.41**
Test time (s)	11.1	8.22	22.8	12.23	10.13	14.75	1.16	4.64	**8.21**
Params(M)	0.08188	0.07032	0.50580	0.51766	0.48442	0.92711	0.38985	0.67903	**0.01983**
FLOPs(G)	0.01066	0.01364	0.02482	0.01781	0.00491	0.04541	8.36811	0.018309	**0.00091**

TABLE III

CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE WHU-LK DATASET WITH 0.5% TRAINING SAMPLES

Class	M3D-DCNN	3D-CNN	DFFN	AB-LSTM	SSFTT	GAHT	DSGSF	GraghGST	Proposed
1	99.46 ± 0.38	99.34 ± 0.14	99.26 ± 0.58	97.55 ± 2.83	99.77 ± 0.14	99.76 ± 0.08	99.58±0.06	98.19±0.32	**99.93 ± 0.03**
2	97.41 ± 0.73	91.40 ± 2.70	93.49 ± 2.57	88.84 ± 4.08	97.68 ± 1.21	98.10 ± 0.94	97.88±1.70	91.07±6.69	**99.77 ± 0.13**
3	90.70 ± 3.94	77.36 ± 3.43	91.16 ± 6.42	60.28 ± 31.45	96.57 ± 1.81	96.93 ± 3.32	97.18±3.10	96.54±0.82	**99.11 ± 0.47**
4	98.97 ± 0.44	98.96 ± 0.28	98.98 ± 0.53	97.92 ± 0.40	99.30 ± 0.13	99.39 ± 0.37	98.79±0.74	97.24±1.46	**99.75 ± 0.09**
5	88.55 ± 2.04	66.84 ± 13.79	78.96 ± 9.58	61.65 ± 29.28	90.75 ± 2.70	92.83 ± 2.30	96.07±1.32	86.61±3.49	**97.93 ± 0.55**
6	99.61 ± 0.35	99.00 ± 0.78	98.32 ± 1.12	92.39 ± 3.60	99.70 ± 0.26	99.67 ± 0.34	99.50±0.24	98.46±0.96	**99.74 ± 0.25**
7	99.97 ± 0.02	99.96 ± 0.03	99.84 ± 0.15	99.97 ± 0.02	**99.98 ± 0.01**	99.89 ± 0.10	99.89±0.08	99.93±0.03	99.96 ± 0.02
8	**96.23 ± 1.28**	94.36 ± 2.01	93.44 ± 0.92	89.36 ± 4.82	94.74 ± 0.89	95.21 ± 0.62	95.93±2.18	96.18±1.09	95.63 ± 1.46
9	94.06 ± 2.82	88.38 ± 5.12	86.66 ± 9.00	57.42 ± 19.05	94.55 ± 1.64	95.84 ± 1.89	**98.13±0.79**	91.72±6.61	96.15 ± 0.82
OA(%)	98.80 ± 0.18	97.64 ± 0.47	98.02 ± 0.54	95.21 ± 2.34	99.07 ± 0.07	99.18 ± 0.12	99.09 ± 0.25	97.69±0.75	**99.57 ± 0.04**
AA(%)	96.11 ± 0.52	90.62 ± 2.43	93.34 ± 2.08	82.82 ± 9.57	97.00 ± 0.28	97.52 ± 0.43	98.11 ± 0.33	95.11±1.04	**98.66 ± 0.16**
K×100	98.42 ± 0.24	96.89 ± 0.63	97.39 ± 0.71	93.67 ± 3.12	98.77 ± 0.09	98.92 ± 0.15	98.81 ±0.32	96.96±0.99	**99.43 ± 0.05**
Train time	369.54	329.42	237.69	232.68	209.83	257.22	817.36	140.49	**101.89**
Test time	36.13	25.31	32.87	26.52	21.76	30.58	1.23	26.19	**11.02**
Params(M)	0.21091	0.12702	0.52980	2.76164	1.25395	1.51412	0.41119	2.91954	**0.01983**
FLOPs(G)	0.03009	0.03571	0.02600	0.06292	0.01287	0.07410	8.87002	0.06439	**0.00091**

TABLE IV

CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE SA DATASET WITH 1% TRAINING SAMPLES

Class	M3D-DCNN	3D-CNN	DFFN	AB-LSTM	SSFTT	GAHT	DSGSF	GraghGST	Proposed
1	98.95 ± 1.56	96.50 ± 2.50	97.65 ± 2.82	97.94 ± 0.77	99.92 ± 0.07	100.00 ± 0.00	99.15±1.47	99.96±0.08	**100.00 ± 0.00**
2	99.41 ± 0.80	99.91 ± 0.08	99.76 ± 0.09	98.56 ± 0.91	99.86 ± 0.21	99.87 ± 0.25	99.54±0.44	98.50±1.13	**99.99 ± 0.02**
3	98.32 ±1.30	92.51 ± 3.58	94.52 ± 3.65	94.11 ± 4.63	97.51 ± 2.57	99.13 ± 0.51	99.42±0.71	99.68±0.31	**100.00 ± 0.00**
4	98.93 ± 0.75	97.67 ± 0.87	97.92 ± 1.28	99.25 ±0.44	99.00 ± 0.65	98.98 ± 1.27	98.83±0.38	98.92±0.69	**99.50 ± 0.32**
5	97.90 ± 0.42	96.03 ± 1.21	96.74 ± 1.45	96.67 ± 1.07	95.84 ± 2.25	98.77 ± 1.37	97.35±2.64	95.65±2.69	**99.05 ± 0.87**
6	99.87 ± 0.19	99.85 ± 0.12	99.99 ± 0.01	98.27 ± 1.64	99.27 ± 1.42	**99.92 ± 0.16**	99.80±0.10	99.54±0.37	99.91 ± 0.15
7	99.66 ± 0.15	99.24 ± 0.23	99.45 ± 0.21	98.61 ± 0.55	99.32 ± 0.18	99.95 ± 0.05	98.75±1.08	99.94±0.05	**99.98 ± 0.02**
8	88.56 ± 2.39	83.78 ± 2.74	90.08 ± 2.60	80.56 ± 5.92	89.69 ± 2.25	91.62 ± 4.22	**99.45±0.22**	95.18±2.52	99.02 ± 0.47
9	99.70 ±0.21	98.31 ± 1.29	97.68 ± 3.49	98.44 ± 1.42	99.75 ± 0.15	99.84 ± 0.18	98.90±0.82	99.37±0.41	**100.00 ± 0.00**
10	91.34 ± 1.04	90.59 ± 1.41	94.48 ± 0.68	86.51 ± 1.99	95.55 ± 0.63	97.49 ± 1.57	**99.06±0.41**	96.05±1.08	98.18 ± 1.24
11	96.10 ± 2.64	86.59 ± 6.13	97.69 ± 2.10	64.16 ± 34.38	94.38 ± 2.40	99.14 ± 0.49	99.23±1.28	96.81±1.85	**99.64 ± 0.20**
12	99.00 ± 0.66	99.08 ± 1.19	**99.99 ±0.02**	97.08 ± 1.33	99.86 ± 0.14	99.61 ± 0.63	98.80±1.08	98.97±0.59	99.98 ± 0.03
13	98.84 ± 0.81	97.55 ± 2.31	**99.64 ± 0.61**	97.71 ± 1.12	97.62 ± 2.86	98.29 ± 2.15	99.16±1.45	98.75±1.53	99.31 ± 0.65
14	95.00 ± 1.11	96.82 ± 1.92	97.50 ± 2.50	85.30 ± 5.01	93.96 ± 2.92	98.68 ± 0.51	98.03±1.09	95.48±2.22	**98.89 ± 0.39**
15	76.99 ± 4.75	75.65 ± 3.12	85.35 ± 4.63	71.69 ± 11.97	79.82 ± 2.80	92.07 ± 5.45	**99.47±0.41**	82.81±3.67	98.92 ± 0.75
16	95.75 ± 2.91	93.30 ± 1.06	97.59 ± 1.55	96.54 ± 1.68	95.06 ± 2.58	98.39 ± 0.83	**99.47±0.82**	97.45±1.20	98.89 ± 0.83
OA(%)	93.29 ± 0.40	91.24 ± 0.56	94.63 ± 0.44	89.11 ± 2.25	93.97 ± 0.31	96.74 ± 0.68	99.15 ± 0.29	95.41±0.59	**99.39 ± 0.10**
AA(%)	95.90 ± 0.28	93.96 ± 0.74	96.63 ± 0.42	91.34 ± 1.79	96.03 ± 0.38	98.23 ± 0.27	99.03 ± 0.37	97.07±0.24	**99.45 ± 0.11**
K×100	92.53 ± 0.45	90.24 ± 0.61	94.02 ± 0.49	87.87 ± 2.51	93.28 ± 0.35	96.38 ± 0.75	99.05 ± 0.32	95.41±0.59	**99.33 ± 0.11**
Train time	159.67	110.12	103.03	101.34	91.44	100.14	274.38	70.2	**66.70**
Test time	14.5	7.11	9.91	8.19	7.49	9.98	0.69	4.37	**4.55**
Params(M)	0.27312	0.16168	0.42336	1.63572	0.95028	1.02562	0.40353	1.45229	**0.01991**
FLOPs(G)	0.02257	0.02702	0.02075	0.04270	0.00972	0.05020	4.54834	0.05977	**0.00091**

tion results on the PU and SA datasets, misclassified pixels are mainly concentrated in the upper part of the images. On the WHU-LK dataset, some pixels in the middle of the image are misclassified. Specifically, on the four datasets, UFMS-LN exhibits significant advantages in classification performance due to its ability to extract global and local features effectively. In conclusion, under limited training sample availability, UFMS-LN demonstrates the best visual classification performance compared to other methods.

In order to compare the performance of the various methods more intuitively, we visualized the features on the SA dataset using t-SNE, as shown in Fig. 12. Observing the results in Fig. 12, the following conclusions can be drawn: the interclass differences between M3D-DCNN and 3D-CNN methods are small and there are a large number of confusing samples between different classes. In contrast, the interclass differences between DSGSF, GAHT, and UFMS-LN methods are larger, while the classification distributions between DSGSF and UFMS-LN methods are similar, but the number of confounded samples in the DSGSF method is more than that in the UFMS-LN method.

E. Time and Complexity Analysis

It is necessary to compare its Params, FLOPs, and time complexity with the state-of-the-art approaches. From Tables II–V,

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TABLE V
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE IP DATASET WITH 3% TRAINING SAMPLES

Class	M3D-DCNN	3D-CNN	DFFN	AB-LSTM	SSFTT	GAHT	DSGSF	GraghGST	Proposed
1	96.36±7.27	93.40±8.89	77.05±12.97	13.28±13.10	97.00±3.32	98.23±1.44	85.26+-11.18	77.79±22.26	**97.63±2.25**
2	65.29±11.25	62.62±5.11	58.30±8.08	70.62±6.59	79.90±5.04	86.40±3.67	87.07+-6.93	**92.60±3.15**	87.81±7.10
3	60.34±9.42	53.94±5.14	51.66±12.84	12.25±16.20	76.69±3.53	76.59±11.20	89.59+-4.68	**95.32±2.65**	91.46±1.54
4	64.91±15.87	50.00±9.72	53.57±14.75	15.63±17.50	67.50±14.82	85.09±5.23	78.23+-16.65	**89.38±5.19**	89.20±5.25
5	84.55±4.49	79.07±4.47	45.26±25.59	22.16±26.07	86.27±2.09	86.37±5.31	92.07+-6.70	**95.13±2.76**	84.94±1.76
6	95.09±3.09	94.83±3.03	90.03±4.31	86.71±7.49	97.43±2.77	98.43±1.39	96.28+-2.95	97.20±2.60	**97.91±1.41**
7	**100.00±0.00**	**100.00±0.00**	93.64±7.32	71.83±28.93	**100.00±0.00**	**100.00±0.00**	48.04+-15.92	50.86±14.88	**100.00±0.00**
8	98.30±0.83	95.98±2.51	93.62±4.55	90.04±3.84	96.72±3.67	99.74±0.25	98.63+-1.70	99.18±0.82	**99.43±1.14**
9	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	21.67±27.91	**100.00±0.00**	**100.00±0.00**	72.51+-20.67	67.48±18.18	**100.00±0.00**
10	71.34±7.51	60.92±2.91	57.54±16.32	13.11±20.81	68.90±6.68	83.74±3.46	89.91+-4.94	**91.68±3.53**	85.80±4.35
11	85.59±5.80	79.98±5.83	81.33±2.22	78.29±4.30	81.72±5.05	87.92±2.60	94.1+-1.67	**94.87±1.92**	93.89±2.87
12	59.57±6.63	47.06±8.26	22.93±11.44	12.86±16.51	65.64±6.78	77.52±4.70	82.99+-7.95	88.50±4.81	**91.99±4.03**
13	99.38±0.39	98.65±0.97	98.33±1.52	35.73±43.34	99.17±0.78	96.98±1.66	95.95+-4.17	**99.59±0.82**	99.06±0.77
14	94.32±1.96	93.94±3.42	93.15±4.74	96.23±2.12	94.63±4.33	92.74±2.24	**98.13+-1.25**	96.66±3.12	97.45±2.52
15	68.99±13.10	65.86±6.08	37.12±9.84	3.24±3.07	66.07±9.04	74.71±9.45	91.95+-3.58	**92.58±1.48**	81.20±12.20
16	99.75±0.50	**100.00±0.00**	**100.00±0.00**	98.75±1.37	99.25±1.00	99.50±0.61	86.12+-5.88	84.87±10.20	98.75±1.37
OA（%）	79.49±4.01	74.61±1.65	69.18±3.00	57.32±2.13	82.21±2.32	87.38±0.78	91.52±2.86	**93.79±0.54**	92.11±1.42
AA（%）	83.99±3.10	79.77±1.14	72.10±3.79	46.40±2.97	86.06±1.66	90.25±0.54	86.68±3.70	88.36±2.38	**93.53±0.88**
K×100	76.43±4.61	70.81±1.77	64.45±3.65	50.06±2.60	79.69±2.63	85.59±0.90	90.31±3.28	**92.92±0.60**	90.98±1.62
Train time (s)	69.79	67.56	76.61	54.96	45.39 seconds	56.63	249.35	**39.11**	41.17
Test time (s)	2.13	1.78	2.4	1.58	1.47 seconds	1.83	0.28	**0.88**	0.94
Params(M)	0.26902	0.16168	0.52022	1.57751	0.93185	1.03816	0.40370	1.03482	**0.01975**
FLOPs(G)	0.02221	0.0265	0.025509	0.04157	0.00953	0.05083	0.89926	0.04369	**0.0009**

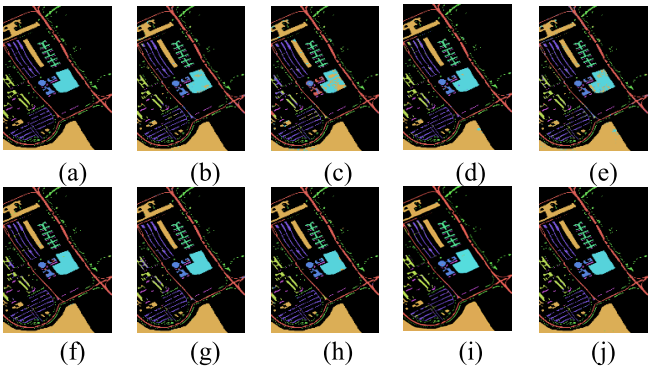

Fig. 10. Classification maps of the PU dataset with 1% training samples. (a) Ground truth. (b) M3D-DCNN. (c) 3D-CNN. (d) DFFN. (e) AB-LSTM. (f) SSFTT. (g) GAHT. (h) DSGSF. (i) GraphGST. (j) Proposed UFMS-LN.

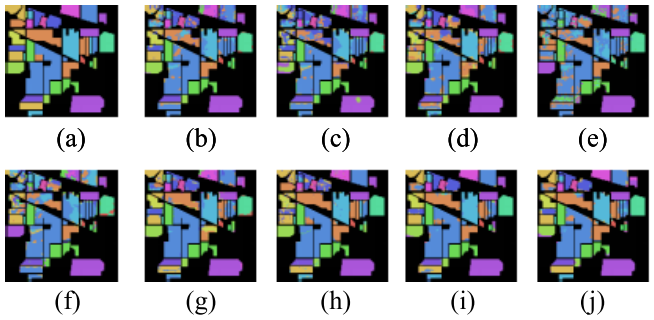

Fig. 11. Classification maps of the IP dataset with 3% training samples. (a) Ground truth. (b) M3D-DCNN. (c) 3D-CNN. (d) DFFN. (e) AB-LSTM. (f) SSFTT. (g) GAHT. (h) DSGSF. (i) GraphGST. (j) Proposed UFMS-LN.

it can be observed that UFMS-LN demonstrates an absolute advantage in terms of both time and computational complexity. In terms of training time, UFMS-LN outperforms other comparison models in most cases on all four datasets, especially demonstrating its lightweight characteristics on the WHU-LK dataset with a large amount of data. CNNs-based models like M3D-DCNN and 3D-CNN require more time on the WHU-LK and SA datasets compared to vision transformer-based models but perform relatively well and require less time on the PU dataset. AB-LSTM falls in the middle range in terms of time consumption on all four datasets. DSGSF requires the longest training time, surpassing other models on all four datasets. However, we can find that although DSGSF has high training time requirements, it has very few testing time requirements. This is due to the fact that the structural design of the DSGSF model requires a smaller volume of data to be involved in the testing and, therefore, a shorter testing time. Notably, GraphGST also has excellent results in terms of runtime speed.

Tables II–V present the average training and testing time, but it does not include the preprocessing stage. The total time of different models is shown in Fig. 13, including preprocessing, training, testing, and other stages. It can be observed that UFMS-LN maintains a lower total time consumption than other models.

In terms of complexity, AB-LSTM exhibits relatively high complexity on all four datasets. SSFTT, GAHT, and GraphGST, which belong to vision transformer-based models, have relatively high parameter counts on four datasets. Additionally, GAHT exhibits high computational complexity in terms of FLOPs on all four datasets. On the other hand, our proposed UFMS-LN demonstrates significant advantages in both parameter and computational complexity compared to other models. This is because the core operator, CF-MHSA, has low complexity itself, and the network structure is kept concise, avoiding excessive stacking of modules. In addition, excellent performance is achieved while keeping complexity low.

F. Ablation Experiments

For the ablation experiments, experiments were conducted to analyze the effects of different modules, as shown in Table VI. Additionally, we designed experiments to replace the self-attention module with other advanced self-attention modules, aiming to compare the complexity of self-attention modules and their contribution to UFMS-LN, as demonstrated in Table VII.

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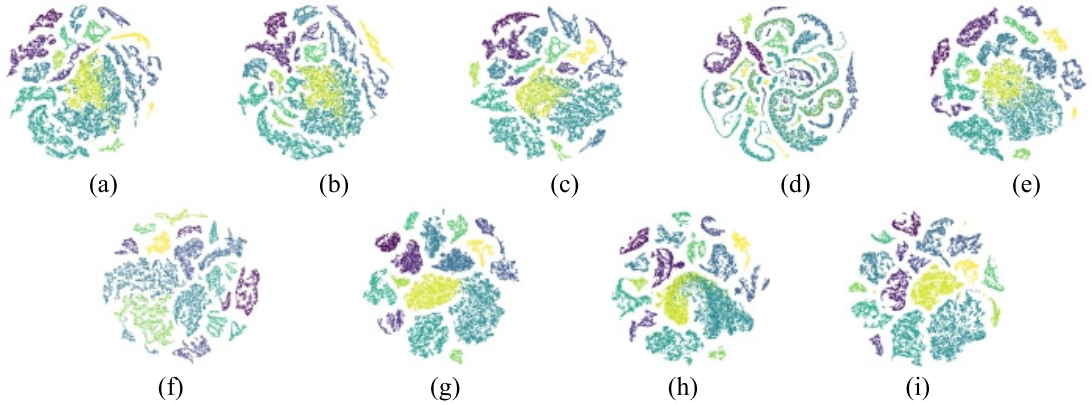

Fig. 12. Visualization with t-SNE on the SA dataset. (a) M3D-DCNN. (b) 3D-CNN. (c) DFFN. (d) AB-LSTM. (e) SSFTT. (f) GAHT. (g) DSGSF. (h) GraphGST. (i) Proposed UFMS-LN.

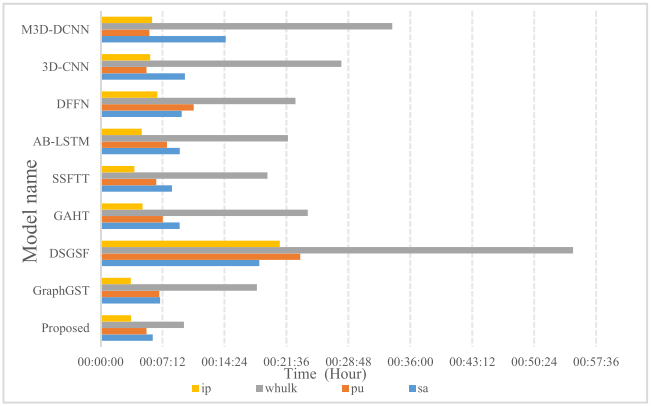

Fig. 13. Operational efficiency of different models on four datasets.

TABLE VI

OAS (%) OF THE ABLATION ON THE UFMS-LN

Module	SA	PU	WHU-LK	IP
CF-MHSA +SH-RFAConv	97.49	98.18	99.46	88.83
SFE-ETR+SH-RFAConv	99.08	98.91	99.11	90.27
CF-MHSA+SFE-ETR	98.47	98.69	98.47	90.28
SH-RFAConv	97.01	98.36	99.47	88.11
CF-MHSA	93.94	97.59	99.20	88.84
SFE-ETR	98.65	98.69	99.30	87.87
CF-MHSA+ETR+SH-RFAConv	98.97	98.61	99.42	87.30
CF-MHSA+SFE-ETR+RFAConv	99.30	99.15	99.43	90.33
CF-MHSA+SFE-ETR+SH-RFAConv	**99.39**	**99.30**	**99.57**	**92.11**

TABLE VII

COMPARISON OF COMPLEXITY AND OAS (%) OF DIFFERENT ATTENTION

Module	MHSA	FousedLiear Attention	Window Attention	Efficient Attention	CF-MHSA
Params	0.01464M	0.01495M	0.01464M	0.00799M	**0.00006M**
FLOPs	0.03528 G	0.03896G	0.03707G	0.01911 G	**0.00011G**
SA	98.92±0.28	98.90±0.13	98.99±0.22	99.00±0.27	**99.39±0.10**
PU	98.85±0.12	98.69±0.12	98.76±0.32	99.11±0.19	**99.30±0.14**
WHU-LK	99.40±0.07	99.29±0.11	99.43±0.14	99.39±0.16	**99.57±0.04**
IP	90.57±0.30	87.94±2.06	90.45±1.09	89.99±1.10	**92.11±1.42**

1) Ablation for SFE-ETR: SFE-ETR serves as the preprocessing module in our network, aiming to reduce the dimensionality of HSI, decrease interclass differences, and increase intergroup distances. To validate the contribution of this module, we compared the performance with and without using it on different datasets, as shown in Table VI.

The experiments indicate that using this module leads to significantly improved classification accuracy compared to not using it. Particularly on the SA dataset, using this module results in a 1.9% increase in OA. This demonstrates the role of SFE-ETR in enhancing features and positively influencing the subsequent classification process. Furthermore, the classification accuracy decreased when SFE-ETR was replaced with ETR. The reason why SFE-ETR is superior to ETR in terms of feature extraction capabilities is that it effectively makes up for the spatial features lost during the dimensionality reduction process.

2) Ablation for CF-MHSA: CF-MHSA plays a crucial role in extracting long-range features. To validate the contribution of this module, we compared the classification accuracy by removing this module and replacing it with other advanced attention mechanisms on different datasets, as shown in Tables VI and VII.

As shown in Table VI, removing the CF-MHSA module results in a certain degree of decrease in the OA metric. This reflects the rationality of designing CF-MHSA.

Furthermore, we propose CF-MHSA as an extremely lightweight self-attention module. Therefore, it is crucial to compare and analyze parameters, complexity, and contribution to the network in comparison with other self-attention modules.

To further demonstrate the superiority of CF-MHSA, we compared it with various advanced self-attention modules, including multihead attention [24], focused linear attention [31], window attention [32], and efficient attention [33]. As shown in Table VII, CF-MHSA consistently outperforms the comparison models in terms of Params and FLOPs by two to four orders of magnitude. CF-MHSA achieves its lightweight volume and computational complexity due to the following factors: 1) feature compression, reducing the number of parameters involved in computations by decomposing the feature matrix into vectors; 2) conversion of matrix operations to vector operations, significantly reducing computational complexity; and 3) the order of operations between query,

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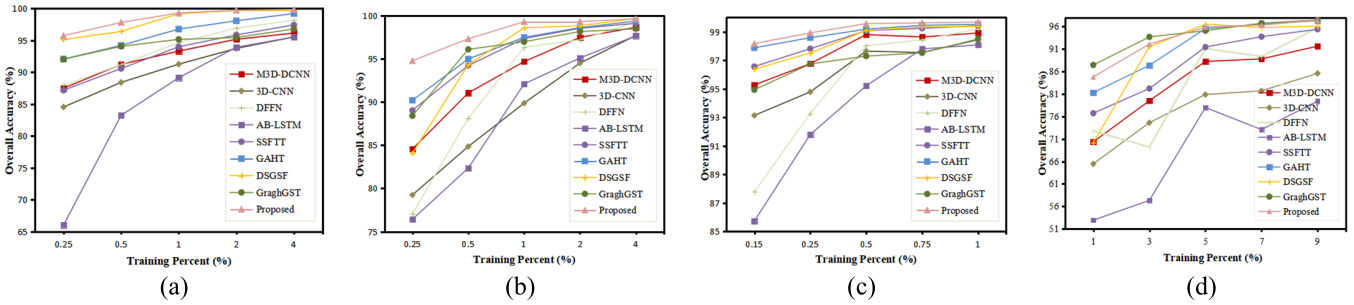

Fig. 14. OAs (%) of UFMS-LN with different training sample ratios on four datasets. (a) SA. (b) PU. (c) WHU-LK. (d) IP.

key, and value is changed, effectively reducing computational complexity.

In terms of parameter count, focused linear attention has a higher Params, specifically 0.014952M, while MHSA and window attention are similar. Regarding computational complexity, focused linear attention has the highest value, 0.038955G. Window attention and attention are similar, with values of 0.037069G and 0.03528G, respectively. On the other hand, efficient attention has lower computational complexity, only 0.01911G.

Overall, CF-MHSA demonstrates significant advantages in terms of parameter count and computational complexity. It has a parameter count of only 0.00006M and a computational complexity of 0.00011G. Despite having fewer parameters and lower computational complexity, CF-MHSA exhibits superior contribution to the model compared to replacing it with other self-attention mechanisms.

3) Ablation Study on SH-RFAConv: The role of SH-RFAConv in this model is to extract local spatial features. To validate the contribution of this module, we compared the performance with and without using SH-RFAConv on different datasets.

The experiments indicate that removing this module leads to a significant decrease in network accuracy, demonstrating the positive effect of SH-RFAConv on UFMS-LN. Additionally, when SH-RFAConv is replaced with RFAConv, the classification accuracy decreases. The reason why SH-RFAConv outperforms RFAConv in terms of feature extraction ability is that it augments the receptive field attention capabilities in both spectral and spatial dimensions, making it more suitable for HSI and allowing for a wider receptive field.

G. Robustness Evaluation

The robustness of a network is crucial in determining its suitability for practical use. To evaluate the robustness of the UFMS-LN, we conducted tests by gradually increasing the training data to observe the network's performance under different training conditions, as shown in Fig. 14. Specifically, 0.25%, 0.5%, 1%, 2%, and 4% of labeled samples from the SA and PU datasets were randomly selected as training samples. For the WHU-LK dataset, the training sample sizes were 0.15%, 0.25%, 0.5%, 0.75%, and 1%. For the IP dataset, we randomly selected 1%, 3%, 5%, 7%, and 9% of the labeled samples as training samples. Average OA was used to describe the accuracy of each model under different training sample sizes. It can be seen that models perform on the four datasets with different numbers of training samples. Compared with other methods, UFMS-LN shows advantageous performance on different datasets. Especially on the PU dataset, UFMS-LN maintains the absolute performance advantage at different training sample rates.

IV. Conclusion

In this article, we propose a UFMS-LN for HSI classification. It primarily consists of CF-MHSA, SH-RFAConv, and SFE-ETR. CF-MHSA is used to extract global features and enable it to capture global contextual information. Its parameter volume and computational complexity can be two orders of magnitude lower than other state-of-the-art attention, truly making it no longer expensive to model long-range dependencies. To complement the local feature extraction capability, we design SH-RFAConv, which has a wider receptive field and can capture features of different dimensions. SFE-ETR acts as a preprocessing module to reduce interclass variations, increase intergroup distances, and highlight features of HSI. UFMS-LN has powerful feature extraction capabilities while achieving lightweight. A series of experiments on four benchmark datasets demonstrate that the proposed UFMS-LN outperforms several state-of-the-art networks in HSI classification.

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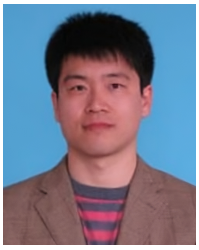

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